

Some review questions - 7th week

Q1 Give the definition of the column space.

Q2 Give the column space of I_n .

Q3 Considering the matrices $A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and the vector $b = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$, determine whether each of the systems $Ax = b$ and $Cx = b$ has a non-empty solution set, and justify your answer using the column spaces of A and C .

Q4 Given a matrix $A \in \mathbb{R}^{p \times n}$ and a vector $b \in \mathbb{R}^p$, prove that the linear system $Ax = b$ admits at least one solution if and only if $b \in C(A)$.

Q5 Express the solution set of the linear system associated with the augmented matrix representation:

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & -1 & 4 & 0 \\ 0 & 1 & 0 & 3 & 1 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Q6 Express the solution set of the system of equations:

$$\begin{cases} x_1 - x_4 + 4x_5 = 1, \\ x_2 + 3x_4 + x_5 = 6, \\ x_3 + 2x_4 + 2x_5 = 7, \\ 0 = c. \end{cases}$$

Q7 Express the following set as the span of two vectors:

$$W = \{[\alpha - 4\beta, -3\alpha - \beta, -2\alpha - 2\beta, \alpha, \beta]^\top \mid \alpha, \beta \in \mathbb{R}\} \subseteq \mathbb{R}^5.$$

Q8 Express the following set as the sum of a vector and the span of two vectors:

$$W = \{[\alpha - 4\beta + 1, -3\alpha - \beta + 6, -2\alpha - 2\beta + 7, \alpha, \beta]^\top \mid \alpha, \beta \in \mathbb{R}\} \subseteq \mathbb{R}^5.$$

Q9 Given a set of k linearly independent vectors $\mathcal{M} = \{u_1, \dots, u_k\}$, does there exist a strictly smaller subset $\mathcal{H} \subsetneq \mathcal{M}$ such that $\text{span}(\mathcal{M}) = \text{span}(\mathcal{H})$? (No proof needed.)

Q10 Given a set of k linearly dependent vectors $\mathcal{M} = \{u_1, \dots, u_k\}$, does there exist a strictly smaller subset $\mathcal{H} \subsetneq \mathcal{M}$ such that $\text{span}(\mathcal{M}) = \text{span}(\mathcal{H})$? (No proof needed.)

Q11 Given two elements u and v in a linear space, show that $\text{span}(u, v, u + v) = \text{span}(u, v)$.

Q12 Give the definition of a basis.

Q13 For each of the sets of vectors defined below, determine whether it forms a basis of $\mathbb{R}^3$, and justify your answer:

$$\mathcal{U} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} \quad \text{and} \quad \mathcal{V} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} \right\},$$

for some $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$.

Q14 Can a linear space have two different bases? If yes, give an example.

Q15 Provide a basis of $\mathbb{R}^2$ and show that it is indeed a basis of $\mathbb{R}^2$.

Q16 Provide a basis for $\mathbb{R}^{2 \times 2}$. No need to justify.

Q17 Determine whether the following statements are true or false (no justification needed):

- (a) If $u_1, u_2, \dots, u_k \in V$ are linearly independent, then they form a basis of V .
- (b) If $u_1, u_2, \dots, u_k \in V$ are linearly independent, then they form a basis of $\text{span}\{u_1, u_2, \dots, u_k\}$.
- (c) If $u_1, u_2, \dots, u_k$ span V , then they form a basis of V .

Q18 Give the definition of the dimension of a linear space.

Q19 Given a linear space V and $u \in V$ such that $u \neq 0$, give a basis for $\text{span}\{u\}$ and provide its dimension.

Q20 Given a basis $\{v_1, \dots, v_n\}$ of a linear space V , what can you say about any family of m elements of V if $m > n$? What can be said about a set of $n + 1$ vectors in $\mathbb{R}^n$?

Q21 Show, using Gauss-Jordan elimination, that the columns of $\begin{bmatrix} 2 & 4 & 6 \\ 1 & 1 & 2 \end{bmatrix}$ are linearly dependent.

Q22 Let V be an n -dimensional vector space, where n is a positive integer. Give two simple ways to check whether a set of n vectors in V is a basis for V .

Q23 Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. State two different properties involving the column space of A that are equivalent to the invertibility of A .

Q24 Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. State two different properties involving linear systems that are equivalent to the invertibility of A .

Q25 Give the definition of the row space and express the row space of the matrix $\begin{bmatrix} 1 & 0 \\ 4 & 3 \\ 2 & 5 \end{bmatrix}$.

Q26 Give the definition of the row rank and the column rank of a matrix. What can be said about these two numbers?

Q27 In the RREF matrix below, prove that there exists a set of three columns that forms a basis of the column space.

$$\begin{bmatrix} 1 & 0 & a & 0 & c \\ 0 & 1 & b & 0 & d \\ 0 & 0 & 0 & 1 & e \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Q28 In the RREF matrix below, prove that there exists a set of three rows that forms a basis of the row space.

$$\begin{bmatrix} 1 & 0 & a & 0 & c \\ 0 & 1 & b & 0 & d \\ 0 & 0 & 0 & 1 & e \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$